

RELEASE NOTES FOR CLINICALGENOME.ORG WEBSITE

(APRIL 7, 2024 UPDATE)

This release of the clinicalgenome.org introduces a number of new features designed to enhance the functionality and usability of the website.

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“QUICK VIEWS” ON TABLE LISTINGS

Nearly every page with a table listing now has expansion row capabilities, also known as “Quick View”, making it easier to see related data without navigating off the current page. Simply look for the big **+** in the first column as shown in the example below and click to expand the row. What additional content is displayed is relative to the context of the current page. Subsequent sections of this document will discuss further the type of content available for viewing.

The screenshot displays the ClinGen Curated Genes table. The table has columns for Gene, Gene Disease Validity, Dosage Sensitivity, Clinical Actionability, Variant Pathogenicity, and Pharmacogenomics. The first column, Gene, contains a large plus sign (+) next to each gene symbol, indicating the expansion row capability. The gene A4GALT (HGNC:18149) is circled in red. The table also includes a search bar, a filter dropdown (None), and a pagination bar showing 1 to 25 of 2878 rows.

Gene	Gene Disease Validity	Dosage Sensitivity	Clinical Actionability	Variant Pathogenicity	Pharmacogenomics
+ A2ML1 HGNC:23336					
+ A4GALT HGNC:18149					
+ AAGAB HGNC:25662					
+ AARS1 HGNC:20					
+ AARS2 HGNC:21022					
+ AASS HGNC:17366					

GNOMAD V4.0 AND ACMG SF V3.2

All relevant presentation of gnomAD and ACMG Secondary Findings information has been updated to versions 4.0 and 3.2 respectively.

CURATED GENES PAGE UPDATES

The Curated Genes Page now has the ability to quickly see the classifications of various curation activities via the Quick View feature. Click on the Expand (+) control to view a summary matrix as shown below:

The screenshot displays the 'Curated Genes' interface. At the top, it shows counts for various categories: 2878 Unique Curated Genes, 2111 Gene-Disease Validity Genes, 1562 Dosage Sensitivity Genes, 286 Actionability Genes, 101 Genes Included on Approved VCEPs, and 129 Pharmacogenomics Genes. Below this is a search bar and a table of genes. The 'ACVRL1' gene (HGNC:175) is expanded, showing a detailed summary matrix.

Curated Genes Summary Matrix for ACVRL1 (HGNC:175):

Gene	Gene Disease Validity	Dosage Sensitivity	Clinical Actionability	Variant Pathogenicity	Pharmacogenomics
ACVRL1 (HGNC:175)	Curated	Curated	Curated	Approved VCEP	

Last Curated Date: 12/05/2022

Click on the disease name, or one of the activity classifications, to view more detailed information about the curation.

ClinGen Curated Disease	MOI	Gene Disease Validity	Dosage Sensitivity	Clinical Actionability	Variant Pathogenicity	Pharmacogenomics
telangiectasia, hereditary hemorrhagic, type 2 MONDO:0010880	AD ⓘ	Definitive	Sufficient	Moderate	ERepo CSpec	
hereditary hemorrhagic telangiectasia MONDO:0019180				Moderate		
				Strong		

NOTE: Dosage Sensitivity also has a non-disease specific score for this gene of 0 (No Evidence) For Triplosensitivity

Below the matrix, other genes are listed: APC (HGNC:583) and ATM (HGNC:795), both with 'Curated' and 'Approved VCEP' status.

Within the summary matrix, you may click on the disease name to navigate to that particular Disease Page, or click on any of the classifications to view the relevant curation details. Clicking on the Collapse (-) control will hide the Quick View and return to the original listing view.

GENE-DISEASE VALIDITY - ALL CURATION PAGE UPDATES

While the basic layout of the Gene-Disease Validity pages has remained consistent for backward compatibility, all have been updated to improve the access to critical information, and to adopt the new ClinGen classification colors for Gene-Disease Validity.

HGNC and MONDO identifiers have also been associated with their respective gene and disease labels. In addition, the Expert Panel name is now a hyperlink.

ClinGen

Clinical Genome Resource

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Gene Symbol

Enter a gene symbol or HGNC ID (Examples: ADNP, HGNC:15766)

Search

Curated Genes

Gene-Disease Validity

Dosage Sensitivity

Clinical Actionability

Curated Variants

Statistics

Downloads

More

?

Gene-Disease Validity

2576

Total Curations

2111

Unique Genes

56

Expert Panels

Click on below to view hidden columns

Search in table

Showing 1 to 25 of 2576 rows

25

rows per page

<

1

2

3

4

5

...

104

>

Gene	Disease	MOI	Expert Panel	SOP	Classification	Last Eval.
+ A2ML1 HGNC:23336	Noonan syndrome MONDO:0018997	AD	RASopathy Gene Curation Expert Panel	SOP5	Disputed	06/07/2018
+ AARS1 HGNC:20	Charcot-Marie-Tooth disease axonal type 2N MONDO:0013212	AD	Charcot-Marie-Tooth Disease Gene Curation Expert Panel	SOP10	Definitive	03/14/2024
+ AARS2 HGNC:21022	mitochondrial disease MONDO:0044970	AR	Mitochondrial Diseases Gene Curation Expert Panel	SOP8	Definitive	04/18/2022
+ AASS HGNC:17366	hyperlysinemia MONDO:0009388	AR	Aminoacidopathy Gene Curation Expert Panel	SOP9	Definitive	10/14/2022
+ ABAT HGNC:23	developmental and epileptic encephalopathy MONDO:0100062	AR	Epilepsy Gene Curation Expert Panel	SOP8	Moderate	04/19/2022
+ ABCA4 HGNC:34	ABCA4-related retinopathy MONDO:0800406	AR	Retina Gene Curation Expert Panel	SOP9	Definitive	10/06/2022
+ ABCB4 HGNC:45	progressive familial intrahepatic cholestasis type 3 MONDO:0011214	AR	General Gene Curation Expert Panel	SOP9	Definitive	11/23/2022
+ ABCB7 HGNC:48	mitochondrial disease MONDO:0044970	XL	Mitochondrial Diseases Gene Curation Expert Panel	SOP9	Moderate	09/28/2023
+ ABCC4 HGNC:55	qualitative platelet defect MONDO:0001197	AR	Hemostasis/Thrombosis Gene Curation Expert Panel	SOP7	Moderate	05/...

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Clicking on the Expert Panel label will open a new browser tab and display the Expert Panel web page, as illustrated below.

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Clinical Domain Working Groups

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Charcot-Marie-Tooth Disease Gene Curation Expert Panel

Affiliated to Neurological Disorders CDWG

[Membership](#)
[Documents](#)

Charcot-Marie-Tooth disease (CMT) is a spectrum of inherited nerve disorders that cause peripheral neuropathy. CMT is regarded as synonymous with “hereditary motor sensory neuropathy” (HMSN). CMT is the most common inherited neurological disorder, affecting approximately 1 in 2500 individuals. Currently, more than 90 genes are reported to cause CMT and are included on clinical genetic testing panels within the Genetic Testing Registry. The goal of the CMT GCEP is to use the ClinGen gene-disease clinical validity framework to evaluate the strength of evidence for these gene-disease associations. These evaluations will enable the development of refined guidelines for clinical diagnosis and management as well as optimize decision making for patients and families.

[View all Gene-Disease Validity classifications approved by this GCEP.](#)

Expert Panel Status - Approved Expert Panel

Step 1	Step 2
Define Group	Expert Panel Approval
Complete	Completed Dec. 2019

Documents

[Conflict Of Interest \(COI\)](#)

ClinGen CMT GCEP COI

[Conflict Of Interest \(COI\)](#) - December 6, 2022

[Charcot-Marie-Tooth Disease Gene Curation Expert Panel](#)

Expert Panel Membership

Membership spans many fields, including genetics, medical, academia, and industry.

Chairs

Chairs

Stephan Züchner, MD, PhD, FAAN

Coordinators

Please contact a coordinator if you have questions.

Dana Bis-Brewer, PhD
dmb107@miami.edu

Gene-Disease Validity Classifications

[View all Gene-Disease Validity classifications approved by this GCEP.](#)

GENE-DISEASE VALIDITY – CURATIONS BY EXPERT PANELS PAGE UPDATES

In the past, clicking on the Expert Panel label in the Curations by Expert Panels Page would navigate to a listing of all curations associated with that expert panel. In this release, the same functionality is provided by the “View Curations” button as shown below.

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Gene Symbol

Enter a gene symbol or HGNC ID (Examples: ADNP, HGNC:15766)

Search

Curated Genes

Gene-Disease Validity

Dosage Sensitivity

Clinical Actionability

Curated Variants

Statistics

Downloads

More

Gene Curation Expert Panels

56

Expert Panels with curations

2644

Total Curations

Click on below to view hidden columns

Search in table

25

rows per page

ClinGen Gene Curation Expert Panel	Primary Curations	Secondary Contributions	View All Curations
+ Aminoacidopathy Gene Curation Expert Panel ClinGen Affiliate ID: 40011	116	-	View Curations
+ Amyotrophic Lateral Sclerosis Spectrum Disorders Gene Curation Expert Panel ClinGen Affiliate ID: 40096	45	2	View Curations
+ Antibody Deficiencies Gene Curation Expert Panel ClinGen Affiliate ID: 40080	65	4	View Curations
+ Arrhythmogenic Right Ventricular Cardiomyopathy Gene Curation Expert Panel ClinGen Affiliate ID: 40003	25	-	View Curations
+ Brain Malformations Gene Curation Expert Panel ClinGen Affiliate ID: 40020	48	-	View Curations
+ Breast/Ovarian Cancer Gene Curation Expert Panel ClinGen Affiliate ID: 40042	29	-	View Curations
+ Brugada Syndrome Gene Curation Expert Panel ClinGen Affiliate ID: 40045	20	-	View Curations
+ Catecholaminergic Polymorphic Ventricular Tachycardia Gene Curation Expert Panel ClinGen Affiliate ID: 40074	12	-	View Curations
+ Cerebral Palsy Gene Curation Expert Panel ClinGen Affiliate ID: 40113	19	-	View Curations

Clicking on the Expert Panel label will now open a new browser tab and display the Expert Panel web page, similar to the behavior of other Gene-Disease Validity pages.

The Curations by Expert Panels Page also gets a new “Quick View”. Clicking on the Expansion (+) control now displays information about the Expert Panel as shown below.

The Dosage Sensitivity All Curations page has always had expansion capabilities, but without a formal control. The capability remains, but now served by Quick View. Clicking on the Expansion (+) control displays additional curation information, including the curated disease as shown below.

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Dosage Sensitivity

Genes: On

Regions: Off

3866

1562

513

Total

Curations

Genes

Regions

Notice:

The standalone region search field specific to Dosage Sensitivity has been fully integrated with the main search bar above. To use, simply select Region (GRCh37) or Region (GRCh38), enter the genomic coordinates as before, then click on the 'Search' button. Once your results display, you can view additional information relative to Dosage Sensitivity by clicking on the 'Dosage Scores' display option.

Advanced Filters: None

Click on below to view hidden columns

25

rows per page

1

2

3

4

5

...

63

Type	Gene Symbol / Region Name	GRCh37	HI Score	TS Score	OMIM Morbid	%HI	pLI	LOEUF	Last Evaluated Date
+ Gene	A4GALT HGNC:18149	22 43088127 43117307	Autosomal Recessive	No Evidence	OMIM Morbid	73.08	0.08	1.39	12/11/2014
- Gene	AAGAB HGNC:25662	15 67493005 67547536	Sufficient Evidence	No Evidence	OMIM Morbid	50.94	0	1.16	02/28/2013
Symbol: AAGAB Previous: No previous names found Curated Loss Disease: palmoplantar keratoderma, punctate type 1A Curated Gain Disease: N/A MONDO									
+ Gene	AARS1 HGNC:20	16 70286201 70323409	No Evidence	No Evidence	OMIM Morbid	24.74	0	0.77	01/11/2018
+ Gene	AARS2 HGNC:21022	6 44266468 44281084	Autosomal Recessive		OMIM Morbid	53.33	0	0.75	08/22/2016
+ Gene	AASS HGNC:17366	7 121713603 121784303	Autosomal Recessive		OMIM Morbid	45.6	0	0.95	08/22/2016

The most significant change in the Dosage Sensitivity All Curations Page is the integration of the formally standalone region search feature with the website’s primary search bar. In previous releases, you had to navigate to the Dosage Sensitivity All Curations Page, then enter the search coordinates to see all a list of genes and regions in various states of curation and pre-curation for Dosage Sensitivity. This same functionality can be easily accomplished through the website’s main search bar. Refer to the “New Search Capabilities” for additional information.

DOSAGE SENSITIVITY – RECURRENT CNV PAGE

The Dosage Sensitivity Recurrent CNV Page gets similar changes to the Dosage Sensitivity All Curations Page described above, including the Quick View Expansion (+) control to show additional curation information as shown below.



Gene Symbol

Search

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Curations of Recurrent CNVs

48
Total
Regions

43
Haplo
Regions

39
Triplo
Regions

[Return to
Dosage Listing](#)

Search in table



Showing 1 to 25 of 48 rows rows per page



< 1 2 >

Region Name	GRCh37	GRCh38	HI Score	TS Score	Last Eval.
+ 1q21.1 recurrent (TAR syndrome) region (proximal, BP2-BP3) (includes RBM8A)	1 145386507 145748064	1 145688228 145748064	Little Evidence	Little Evidence	05/18/2017
+ 1q21.1 recurrent region (BP3-BP4, distal) (includes GJA5)	1 146577486 147394506	1 147105804 147917509	Sufficient Evidence	Sufficient Evidence	04/14/2016
+ 2q11.2 recurrent region (includes ARID5A, TMEM127)	2 96739012 97671429	2 96073264 97005692	Sufficient Evidence	Little Evidence	02/21/2022
+ 2q13 recurrent region (includes NPHP1)	2 110862108 110983703	2 110104531 110228181	Autosomal Recessive	Sensitivity Unlikely	02/25/2016
+ 2q13 recurrent region (includes BCL2L11)	2 111392193 113104742	2 110634616 112347165	Emerging Evidence	Emerging Evidence	06/23/2018
+ 2q21.1 recurrent region (includes ARHGEF4, GPR148)	2 131477509 131929693	2 130719936 131172120	Little Evidence	No Evidence	09/30/2019
+ 3q29 recurrent region (includes DLG1)	3 195756054 197344662	3 196029183 197617791	Sufficient Evidence	Little Evidence	02/18/2024



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Curations of Recurrent CNVs

48

Total Regions

43

Haplo Regions

39

Triplo Regions

[Return to Dosage Listing](#)

Showing 1 to 25 of 48 rows

25 rows per page

1

2

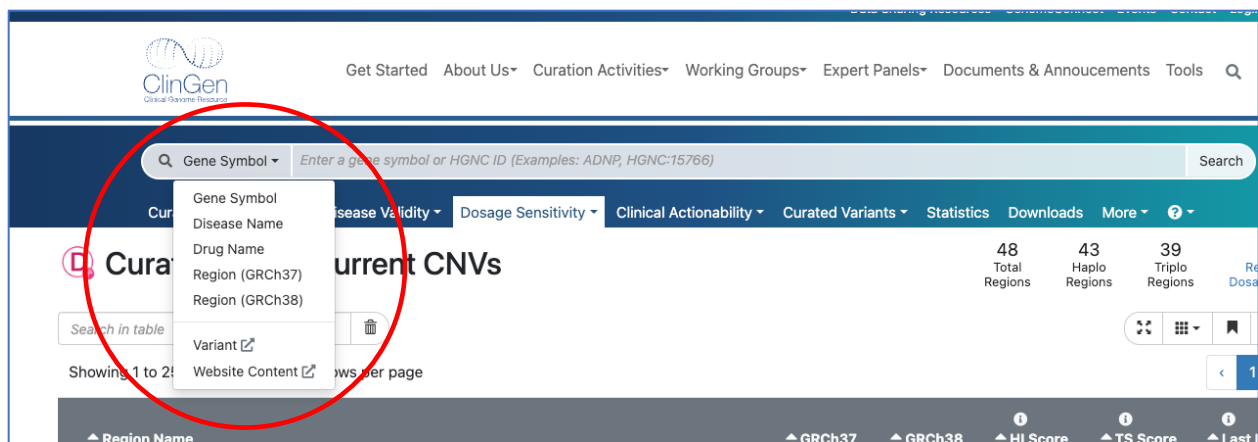
Region Name	GRCh37	GRCh38	HI Score	TS Score	Last Eval.
+ 1q21.1 recurrent (TAR syndrome) region (proximal, BP2-BP3) (includes RBM8A)	1 145386507 145748064	1 146688899 146888899	Little Evidence	Little Evidence	05/18/2017
- 1q21.1 recurrent region (BP3-BP4, distal) (includes GJA5)	1 146577486 147394506	1 147105904 147917509	Sufficient Evidence	Sufficient Evidence	04/14/2016
Region: 1q21.1 recurrent region (BP3-BP4, distal) (includes GJA5) ClinGen DCI ID: ISCA-37421 Curated Loss Disease: chromosome 1q21.1 deletion syndrome Curated Gain Disease: chromosome 1q21.1 duplication syndrome					
+ 2q11.2 recurrent region (includes ARID5A, TMEM127)	2 96739012 97671429	2 96073264 97005692	Sufficient Evidence	Little Evidence	02/21/2022
+ 2q13 recurrent region (includes NPHP1)	2 110862108 110983703	2 110104531 110228181	Autosomal Recessive	Sensitivity Unlikely	02/25/2016
+ 2q13 recurrent region (includes BCL2L11)	2 111392193 113104742	2 110634616 112347165	Emerging Evidence	Emerging Evidence	06/23/2018
+ 2q21.1 recurrent region (includes ARHGEF4, GPR148)	2 131477509 131929693	2 130719936 131172120	Little Evidence	No Evidence	09/30/2019

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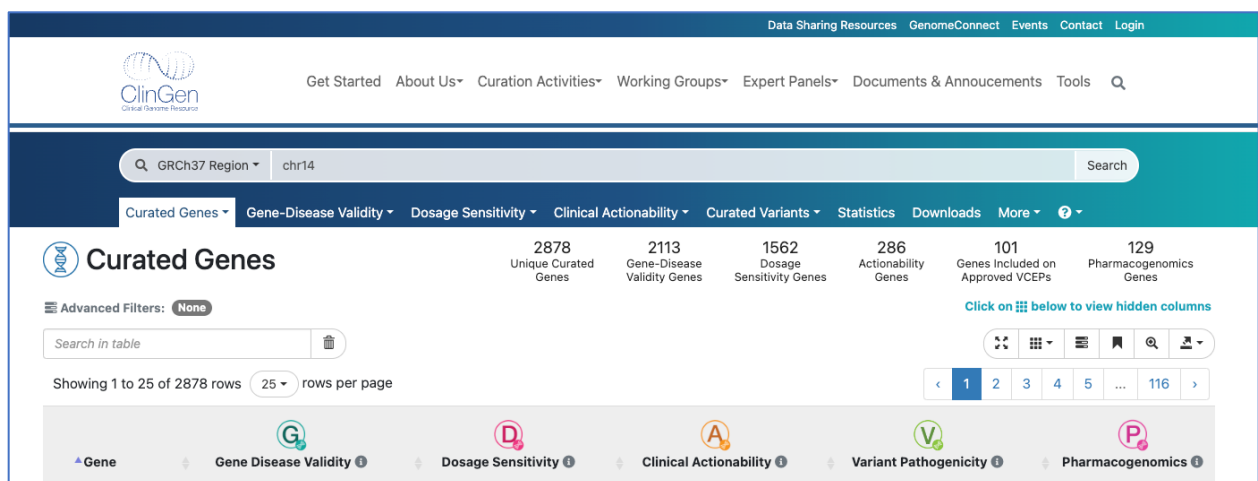
NEW SEARCH CAPABILITIES

The website has always provided the ability to search by gene symbol, disease label, and by GRCh37 or GRCh38 coordinate regions. With this release, both the performance and the display of those search results has greatly improved, particularly with the coordinate region searches.

As before, a coordinate region search begins by selecting either “Region (GRCh37)” or “Region (GRCh38)” from the main search bar menu as shown below.



Once a region build is selected, a suitable coordinate range is entered into the search field, and clicking on the “Search” button (or simply press the return key) begins the search. In the example below, “chr14” was entered to show all activity on chromosome 14.



The results of said search will display as shown below. Note the inclusion of several new elements on this page, including OMIM, Morbid, %HI, pLI, and LOEUF. Similarly, two new columns have been added at the beginning of each row. The first is the Quick View Expansion (+) control as seen on other pages. The second is basic information about the results, including the gene or region type and whether the resultant item is wholly contained within the search coordinates, or overlaps the coordinates.

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Q

Gene Symbol

Enter a gene symbol or HGNC ID (Examples: ADNP, HGNC:15766)

Search

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Location:

chr14

Genes:

1591 total genes

Regions:

21 total regions

Display:

☒ Genes
 ☒ Contained
 ☐ Dosage Scores
 ☒ Pseudogenes

☐ Regions
 ☒ Overlap

Advanced Filters:

None

Click on below to view hidden columns

Showing 1 to 25 of 1591 rows

25 rows per page

<

1

2

3

4

5

...

64

>

Type	Gene / Region	GRCh37	OMIM	%HI	pLI	LOEUF	Curation Activity
+ Gene Contained	SIX6 HGNC:10892	14 60975864 60979568	OMIM Morbid	5.88	0	0.92	Q Last Curated: 03/22/2012
- Gene Contained	MEG3 HGNC:14575	14 101292445 101327363	OMIM	-	-	-	Q Last Curated: 04/25/2012
Last Curated Date: 04/25/2012 Click on the disease name, or one of the activity classifications, to view more detailed information about the curation. ClinGen Curated Disease MOI Q Q A V P paternal uniparental disomy of chromosome 14 MONDO:0011975 NOTE: Dosage Sensitivity also has a non-disease specific score for this gene of 0 (No Evidence) For Triplosensitivity							
+ Gene Contained	DLK1 HGNC:2907	14 101193229 101204561	OMIM	24.61	0.99	0.46	Q Last Curated: 04/28/2012
+ Gene Contained	SIX4 HGNC:10890	14 61176257 61191069	OMIM	18.53	0.01	0.68	Q Last Curated: 05/08

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Within the summary matrix, you may click on the disease name to navigate to that particular Disease Page, or click on any of the classifications to view the relevant curation details. Clicking on the Collapse (-) control will hide the Quick View and return to the original listing view.

Also new to the results page are Display options that can be selected and deselected as desired. With these options, you can show or hide genes, pseudogenes, regions, items fully contained within the search coordinates, items that overlap the search coordinates. You may also choose to display additional information specific to Dosage Sensitivity. In the example below, the Regions option was deselected and the Dosage Sensitivity option selected, showing only genes on the resultant page along with Dosage Sensitivity Haploinsufficiency and Triplosensitivity scores.

With the Dosage Sensitivity Display option selected, the full suite of Dosage Sensitivity Advanced Filters are available as shown below. Without the Dosage Sensitivity Display option selected, the “Show only curated Genes/Regions” filter is still available.

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Enter a gene symbol or HGNC ID (Examples: ADNP, HGNC:15766)

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CHD8
[View Gene Facts](#)

1 Gene-Disease Validity Classifications
 2 Dosage Sensitivity Classifications
 0 Clinical Actionability Assertions
 0 Variant Pathogenicity Assertions
 0/0 CPIC / PharmGKB High Level Records
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GenomeConnect

GenomeConnect partners with individual patients and gene/condition specific registries to share genetic and health data with ClinVar. GenomeConnect participants have the option to connect with other participants, learn about research opportunities, and receive updates about their genetic test results. View the [GenomeConnect Page](#) for more information.

ClinVar Submissions

There are 9 GenomeConnect ClinVar submission(s) for the gene CHD8. [Click here to view](#)

GenomeConnect submits participant genetic and health information to ClinVar as "Phenotyping Only" submissions. These submissions provide additional case-level details and do not count towards the aggregate ClinVar classification. GenomeConnect does not independently classify variants, but shares variant information as it appears on participant reports including the testing laboratory, laboratory classification, and classification date. The submissions also include health information from participant surveys. These details can be viewed by clicking on the down arrow in the 'More Information' column of the 'Submissions-Germline section' for the selected ClinVar variant. Note that GenomeConnect submissions can typically be found near the bottom of the Submissions-Germline list.

Questions?

Email info@genomeconnect.org with questions about GenomeConnect or specific GenomeConnect ClinVar submissions. Additional patient-provided information may be available in the registry. GenomeConnect also has the ability to re-contact participants to request more information.

[Click here to learn more about how to cite ClinGen and to see a list of the external resources utilized.](#)

The information on this website is not intended for direct diagnostic use or medical decision-making without review by a genetics professional. Individuals should not change their health behavior solely on the basis of information contained on this website. If you have questions about the information contained on this website, please see a healthcare professional.

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GenomeConnect is a ClinGen initiative that allows individuals and patient groups to share genetic information to increase the overall understanding of conditions and potential treatments. More information about GenomeConnect is available from its website <https://www.clinicalgenome.org/genomeconnect/>.

Within the new GenomeConnect, along with information content, is a link to the GenomeConnect submissions to ClinVar specific to the relevant gene. Clicking on the link will open a new tab to ClinVar and display the submitted variant entries.

An official website of the United States government [Here's how you know](#)

NIH National Library of Medicine
National Center for Biotechnology Information

ClinVar

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i We've updated the ClinVar website to better support classifications of somatic variants!
Read more about changes to the website in our [web release notes](#); more information about somatic variants in ClinVar is available on [GitHub](#).

Classification type
☐ Germline (9)
☐ Somatic (0)

Germline classification
☐ Conflicting classifications (1)
☐ Benign (0)
☐ Likely benign (0)
☐ Uncertain significance (0)
☐ Likely pathogenic (1)
☐ Pathogenic (5)

Types of conflicts
☐ P/LP vs L/B (0)
☐ P/LP vs VUS (0)
☐ VUS vs L/B (1)

Molecular consequence
☐ Frameshift (2)
☐ Missense (1)
☐ Nonsense (3)
☐ Splice site (1)
☐ ncRNA (0)
☐ Near gene (0)
☐ UTR (0)

Variation type
☐ Deletion (1)
☐ Duplication (2)
☐ Indel (0)
☐ Insertion (1)
☐ Single nucleotide (6)

Variation size
☐ Short variant (< 50 bps) (8)
☐ Structural variant (>= 50 bps) (0)

Search results
Display options Items: 9

Variation	Gene (Protein Change)	Type (Consequence)	Condition	Classification, Review status
☐ NM_001170629.2(CHD8):c.6795C>T (p.Phe2265=)	CHD8, LOC126861888	Single nucleotide variant (synonymous variant)	Inborn genetic diseases +1 more	G Conflicting classifications of pathogenicity ★
☐ NM_001170629.2(CHD8):c.5607dup (p.Asp1870fs)	CHD8 (D1591fs +1 more)	Duplication (frameshift variant)	Inborn genetic diseases	G Pathogenic ★
☐ NM_001170629.2(CHD8):c.4378C>T (p.Arg1467Ter)	CHD8 (R1181* +1 more)	Single nucleotide variant (nonsense)	Intellectual developmental disorder with autism and macrocephaly +1 more	G Pathogenic ★★
☐ NM_001170629.2(CHD8):c.3308G>T (p.Gly1103Val)	CHD8 (G1103V +1 more)	Single nucleotide variant (missense variant)	not provided	G Likely pathogenic ★
☐ NM_001170629.2(CHD8):c.3307+1G>A	CHD8	Single nucleotide variant (splice donor variant)	Autism spectrum disorder +1 more	G not provided ☆
☐ NM_001170629.2(CHD8):c.2854C>T (p.Arg952Ter)	CHD8 (R673* +1 more)	Single nucleotide variant (nonsense)	not provided	G Pathogenic ★
☐ NM_001170629.2(CHD8):c.2345del (p.His782fs)	CHD8 (H503fs +1 more)	Deletion (frameshift variant)	not provided	G Pathogenic ★
☐ NM_001170629.2(CHD8):c.142C>T (p.Gln48Ter)	CHD8 (Q48*)	Single nucleotide variant (nonsense)	not provided	G Pathogenic ★
☐ GRCh37/hg19 14q11.2-21.1(chr14:20511672-42881888)x3	AKAP6, MDP1 +187 more	Copy number gain	not provided	G not provided ☆

CURATION PAGE UPDATES – CLINGEN CURATION ID

Frequently, the clinicalgenome.org user will want to bookmark a page, possibly for inclusion in a future paper. The challenge is that certain pages may have page addresses that are quite long. For example, one of the Gene-Disease Validity curation pages has an address of:

https://clinicalgenome.org/kb/gene-validity/CGGV:assertion_3134b4e1-2fee-4025-99afb2146cc2ccf8-2018-07-16T160000.000Z?page=1&size=25&search=

In the past, users have been forced to use TinyURL or similar services to shorten the address, although this introduces other problems. With the release, the concept of a ClinGen Curation ID is introduced. The ClinGen Curation ID, or CCID for short, is a short identifier that can be used interchangeably with the full page address in any documentation or papers. Unlike using TinyURL, the CCID will always point to the latest released version of a curation, ensuring that references don't become stale and unreachable over time. Currently a CCID is available for Gene-Disease Validity and Dosage Sensitivity curations. Shown below is the new CCID line on a Gene Disease Validity curation page.

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Gene-Disease Validity Details

Gene:
CHD8
HGNC:20153

Definitive
Classification - 07/16/2018

Disease:
complex neurodevelopmental disorder
MONDO:0100038

SOP:
ClinGen Gene Validity Evaluation Criteria SOP5

Mode of Inheritance:
Autosomal dominant inheritance
HP:0000006

Expert Panel:
Intellectual Disability and Autism

Replication over time:
YES

Contradictory Evidence:
NO

ClinGen Curation ID:
CCID:004443
Copy https://search.clinicalgenome.org/CCID:004443 to clipboard

Summary
Genetic Evidence (12.00)
Experimental Evidence (3.50)
Non-Scorable Evidence
Lumping & Splitting
References

CHD8 - complex neurodevelopmental disorder

Evidence Summary:
The CHD8 gene was first reported in relation to autosomal dominant autism spectrum disorder in 2012 (O'Roak et al., PMID: 22495309). At least 8 truncating variants have been reported in 4 publications (PMIDs: 22495309, 24998929, 26789910, 27824329). Most variants are de novo; one individual inherited a CHD8 variant from a heterozygous mother who has some autistic features but has not been clinically diagnosed. More evidence is available in the literature, but the maximum score for genetic evidence (12 points) has been reached. In addition to an increased likelihood of a diagnosis of autism spectrum disorder, individuals with pathogenic variants in CHD8 have variable degrees of intellectual disability, an overgrowth phenotype with macrocephaly and increased height, and dysmorphic facial features (PMID: 24998929). We therefore decided to curate for the generic term complex neurodevelopmental disorder. This gene-disease association is supported by a zebrafish model and a mouse model. In summary, CHD8 is definitively associated with autosomal dominant complex neurodevelopmental disorder. This classification was approved by the ClinGen Intellectual Disability and Autism Gene Curation Expert Panel on 7/16/2018.

To the immediate right of the CCID is a convenient button for copying the CCID page address into the user's clipboard, which can then be easily pasted into any new document or reference.

CLINGEN DASHBOARD UPDATES

With this release, ClinGen Dashboard users can now follow curation activity by disease. Previously, the Dashboard only allowed follow activity by gene. The list of "Followed Disease" has its own expandable header as shown below.

Global notifications are **PAUSED**

GenomeConnect +

Reports +

Followed Genes +

Followed Diseases -

Search in table  Follow New Disease

Name	Curation Status	Last Updated	Notify	Unfollow
All Diseases	    		Default ▾	★
biotinidase deficiency	    	3/23/2024	Default ▾	★
complex hereditary spastic paraplegia	    	11/20/2023	Default ▾	★

Showing 1 to 3 of 3 rows

[Click here to learn more about how to cite ClinGen and to see a list of the external resources utilized.](#)

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ch.clin-gen.info/dashboard#collapseDiseases

Following a disease is very similar to following a gene. Simply click on the “Follow New Disease” button and enter a disease name or MONDO ID as shown below.

Follow a new disease ✕

Disease

My ClinGen Dashboard

Once a disease has been selected, notification frequency can be managed the same as with Follow Genes.

With the addition of a new “Followed” category, and the expectation that more will be offered in the future, it was necessary to alter the controls for globally turning notifications on and off. In the past, a control was present on the “Followed Genes” header. Now, the control is available either through the Settings icon, or by clicking on the new Status Section located above the Reports and Followed headers.

The screenshot displays the ClinGen My ClinGen Dashboard. At the top, there is a navigation bar with links: Data Sharing Resources, GenomeConnect, Events, Contact, and a user profile for Phillip Weller. Below this is a search bar with the placeholder text "Enter a gene symbol or HGNC ID (Examples: ADNP, HGNC:15766)". A dropdown menu for "Curated Genes" is open, showing "All Curated Genes". The main heading is "My ClinGen Dashboard". Below this, a light blue box indicates "Global notifications are ON". Underneath are three colored buttons: "Reports" (blue), "Followed Genes" (green), and "Followed Diseases" (orange), each with a plus icon. To the right is a user profile for Phillip Weller, showing a bio, interests, and a settings gear icon circled in red. At the bottom, there is a footer with a link to learn more about citing ClinGen and a disclaimer.

Global notifications are ON

Reports

Followed Genes

Followed Diseases

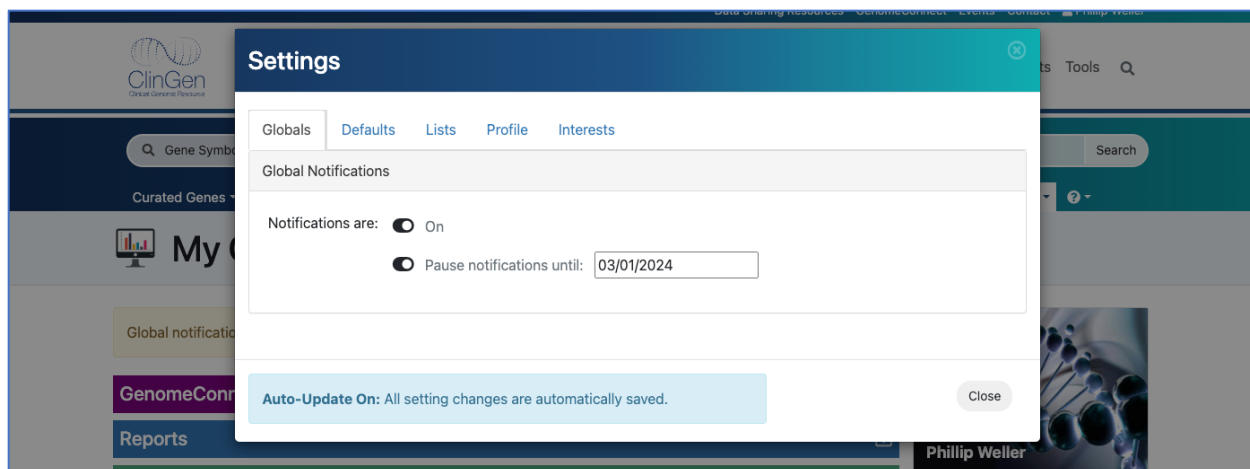
Phillip Weller

Interests:

[Click here to learn more about how to cite ClinGen and to see a list of the external resources utilized.](#)

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Clicking on the Status Section will display the Settings modal where global changes can be made to global notifications.



Besides simply turning Global Notifications on and off, this release also allows a “vacation” type feature, where notifications can be paused until a certain date. The above example shows the feature turned on and a date entered. While a past date cannot be initially entered, once the date has passed, the system will automatically reactivate notifications, even if the Pause feature is still active. The Status Section will reflect any changes made to either global notifications of the vacation feature.

NEW DOWNLOAD REPORTS

For those interested in Dosage Sensitivity curations, a new report has been added to the “Downloads” section. This new report, titled “Dosage Sensitivity Curations (Genes and Regions)” combines all Dosage Sensitivity curations into a single report. Previously genes and regions were only available in separate reports. The separate reports remain, however users importing ClinGen data into their local systems may find the combined report a more convenient vehicle for doing so.

Dosage Sensitivity Downloads

README

This files are produced by the ClinGen Dosage Sensitivity Curation Working Group, formerly the International Standards for Cytogenomic Arrays (ISCA) Consortium Evidence-Based Review Committee. The goal of this group is to curate regions of the genome (both single genes and particular genomic regions) with respect to their haploinsufficiency and/or triplosensitivity.

[Download README](#)

Dosage Sensitivity Curations (Genes Only)

This file provides a summary of the Dosage Sensitivity gene curations completed by ClinGen. This file is built in real-time.

[Download CSV](#)

Dosage Sensitivity Curations (Genes and Regions)

This file provides a summary of the Dosage Sensitivity gene and region curations completed by ClinGen. This file is built in real-time.

[Download CSV](#)

Dosage Sensitivity Curated Gene List

The files in this directory contain data for genes that have been through the review process. These files are updated daily. Files are available for genes and regions localized on both GRCh37 and GRCh38. The tsv files have a header and contain all of the curation information found on the ClinGen Dosage Sensitivity Map web pages, including disease name (when applicable), PMIDs used as evidence, and comments.

[Download GRCh37 \(tsv\)](#)[Download GRCh38 \(tsv\)](#)

Dosage Sensitivity Curated Region List

The files in this directory contain data for regions that have been through the review process. These files are updated daily. Files are available for genes and regions localized on both GRCh37 and GRCh38. The tsv files have a header and contain all of the curation information found on the ClinGen Dosage Sensitivity Map web pages, including disease name (when applicable), PMIDs used as evidence, and comments.

[Download GRCh37 \(tsv\)](#)[Download GRCh38 \(tsv\)](#)

Dosage Sensitivity Haploinsufficiency Genes

BED files are available for gene curation. There are separate files for haploinsufficiency and triplosensitivity as the BED file only has one column for 'score'. Because the score column expects a number rather than text, "Dosage sensitivity unlikely" is represented by the score of 40, and "Gene associated with autosomal recessive phenotype" is represented by the score of 30.

[Download GRCh37 \(bed\)](#)[Download GRCh38 \(bed\)](#)

Dosage Sensitivity Triplosensitivity Genes

BED files are available for gene curation. There are separate files for haploinsufficiency and triplosensitivity as the BED file only has one column for 'score'. Because the score column expects a number rather than text, "Dosage sensitivity unlikely" is represented by the score of 40, and "Gene associated with autosomal recessive phenotype" is represented by the score of 30.

[Download GRCh37 \(bed\)](#)[Download GRCh38 \(bed\)](#)

Dosage Sensitivity Recurrent CNV

These files contains recurrent copy number variations (CNV) that have been, or are in the process of being, reviewed by the recurrent CNV ClinGen DSC Subgroup. This file is available in genome build hg19 and hg38 and can be opened in ChAS software (.aed) or in array analysis software that utilizes (.bed) files as well as the UCSC genome browser.

[Download HG19 \(bed\)](#)[Download HG38 \(bed\)](#)[Download HG19 \(aed\)](#)[Download HG38 \(aed\)](#)

DISEASE FACTS

A widely used feature on the website is something called “Gene Facts”. Gene Facts is present on every gene page, regardless of whether ClinGen has curated that gene or not.

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PLN

View Gene Facts

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0 / 0 CPIC / PharmGKB High Level Records
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Gene-Disease Validity

Group By Activity
Group By Gene-Disease Pair

Gene	Disease	MOI	Expert Panel	Classification	Report & Date
PLN	intrinsic cardiomyopathy MONDO:0000591	AD	Hypertrophic Cardiomyopathy GCEP	Definitive	02/12/2021
PLN	arrhythmogenic right ventricular cardiomyopathy MONDO:0016587	AD	Arrhythmogenic Right Ventricular Cardiomyopathy GCEP	Moderate	12/17/2020

Click here to learn more about how to cite ClinGen and to see a list of the external resources utilized.

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Working Groups

@ClinGenResource
youtube.com
Actionability
Ancestry and Diversity

Clicking on Gene Facts displays a plethora of information accumulated from various sources relative to that gene, as shown below.



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PLN

2

0

0

0

0 / 0

★

Gene-Disease Validity Classifications

Dosage Sensitivity Classifications

Clinical Actionability Assertions

Variant Pathogenicity Assertions

CPIC / PharmGKB High Level Records

Follow Gene

[View Gene Facts](#)

Gene Facts
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HGNC Symbol
 PLN (HGNC:9080)
 [HGNC](#)
[Entrez](#)
[Ensembl](#)
[OMIM](#)
[UCSC](#)
[Uniprot](#)
[GeneReviews](#)
[LOVD LSDB](#)
[ClinVar](#)

HGNC Name
 phospholamban

Gene type
 protein-coding gene

Locus type
 gene with protein product

Previous symbols
 PLB

Alias symbols
 CMD1P

GenCC Classifications
Strong 4
Limited 1
Supportive 1
Moderate 1
Definitive 2
[\(Read more about GenCC Classifications\)](#)

%HI
 10.26
 [\(Read more about the DECIPHER Haploinsufficiency Index\)](#)

pLI
 0.45
 [\(Read more about gnomAD v4.0 pLI score\)](#)

LOEUF
 1.56
 [\(Read more about gnomAD v4.0 LOEUF score\)](#)

pLI and LOEUF metrics based on transcript

ACMG SF v3.2 Gene?
 No

Cytoband
 6q22.31

Genomic Coordinates
[GRCh37/hg19](#): chr6:118869459-118882879
 [NCBI](#)
[Ensembl](#)
[UCSC](#)

[GRCh38/hg38](#): chr6:118548296-118561716
 [NCBI](#)
[Ensembl](#)
[UCSC](#)

MANE Select Transcript
 NM_002667.5
 ENST00000357525.6
 [\(Read more about MANE Select\)](#)

Function
 Reversibly inhibits the activity of ATP2A2 in cardiac sarcoplasmic reticulum by decreasing the apparent affinity of the ATPase for Ca(2+) (PubMed:28890335). Modulates the contractility of the heart muscle in response to physiological stimuli via its effects on ATP2A2. Modulates calcium re-uptake during muscle relaxation and plays an important role in calcium homeostasis in the heart muscle. The degree of ATP2A2 inhibition depends on the oligomeric state of PLN. ATP2A2 inhibition is alleviated by... (Source: Uniprot)

[Curation Summaries](#)
[Status and Future Work 2](#)
[External Genomic Resources](#)
[ClinVar Variants](#)


Gene-Disease Validity

[Group By Activity](#)
[Group By Gene-Disease Pair](#)

With this release, ClinGen is applying the same resource to diseases, known as Disease Facts. While the scope of information is not yet as significant as applied to genes, Disease Facts provides a useful source of accumulated information relative to the viewed disease. The example below shows Disease Facts expanded for the disease “intrinsic cardiomyopathy”.

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intrinsic cardiomyopathy

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Dosage Sensitivity Classifications

0

Clinical Actionability Assertions

0

Variant Pathogenicity Assertions

Disease Facts

External Data Attribution

Name

intrinsic cardiomyopathy

MONDO

Ontological Reference

MONDO:0000591

Description

A cardiomyopathy that is due to abnormalities in heart muscle cells. (Source: MONDO)

Synonyms

intrinsic cardiomyopathy
primary cardiomyopathy

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Group By Activity

Group By Gene-Disease Pair

Gene	Disease	MOI	Expert Panel	Classification	Report & Date
ACTN2	intrinsic cardiomyopathy	AD	Hereditary Cardiovascular Disease GCEP	Definitive	06/10/2023
PLN	intrinsic cardiomyopathy	AD	Hypertrophic Cardiomyopathy GCEP	Definitive	02/12/2021

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